

RUSSELL HEAD

Technical Solutions Architect · Senior Systems & Automation Engineer
Sovereign AI · Predictive Automation · Bare-Metal Infrastructure

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EXECUTIVE SUMMARY

Solutions architect and senior systems engineer with 17+ years shipping production infrastructure, predictive automation, and sovereign AI systems end-to-end. At Amazon UK, conceived and built the logistics forecasting framework that captured ~£400k in recurring annual savings — adopted facility-wide. As founder of a 17-year engineering consultancy, replaced variable-cost commercial LLM APIs with self-hosted Ollama environments and custom Python agents, eliminating AI spend and recovering full data sovereignty. Bare-metal Linux specialist with a track record of mentorship at Amazon and a consistent record of translating ambiguous stakeholder requirements into phased, measurable technical roadmaps.

KEY IMPACT

£400,000 Recurring annual savings — Amazon UK logistics forecasting framework, adopted facility-wide	100% Elimination of variable AI inference spend via self-hosted Ollama + custom Python agents
30+ Production web systems architected, built, and maintained across 17 years of consultancy	500,000+ Organic international views via automated asset-rendering and multi-platform distribution

CORE COMPETENCIES

Sovereign AI & LLMOps	Self-hosted LLMs (Ollama, llama.cpp, GGUF/AWQ/GPTQ) · RAG pipelines · Vector DBs (Chroma, Qdrant, FAISS) · Embedding pipelines · Multi-model orchestration · Inference optimisation · Custom Python agents · Prompt engineering
Systems & Infrastructure	Bare-metal Linux (Ubuntu Server, Linux Mint) · systemd · Nginx · Docker/Podman · ZeroTier VPN · Kernel & I/O tuning · iptables/UFW · Observability (Prometheus/Grafana) · Bash & Python automation
Full-Stack Engineering	Python · JavaScript (ES6+) · PHP · MySQL · HTML5/CSS3 · Web Audio API · LSL (Linden Scripting Language)
Performance & Growth	Technical SEO · Core Web Vitals · JSON-LD schema · Semantic HTML5 · Asset delivery & caching · Crawl-budget engineering
Leadership	Stakeholder translation · Phased roadmap delivery · Technical mentorship · Workshop facilitation · Engineering handoffs

PROFESSIONAL EXPERIENCE

Founder & Technical Director

Russell Head Digital — Luton, UK · 2008 – Present

End-to-end engineering consultancy. Sole principal across full lifecycle — from ambiguous stakeholder requirements through architecture, build, deployment, and ongoing operations. 17 years, 30+ production systems.

Sovereign AI Platform

- Replaced variable-cost commercial LLM APIs with self-hosted Ollama environments orchestrated by custom Python agents — cutting AI inference spend to zero, recovering full data privacy, and making infrastructure spend fully predictable.
- Designed RAG pipelines over private corpora using vector databases (Chroma / Qdrant / FAISS), embedding models, and prompt orchestration. Tuned GGUF/AWQ/GPTQ quantisation and CPU/GPU allocation for sub-second local inference.
- Built multi-step, tool-using autonomous agents replacing SaaS workflows with deterministic, auditable local execution — including predictive bottleneck detection in operational pipelines.

Full-Stack & Browser Engineering

- Architected, built, and deployed 30+ production web systems, custom database schemas, and browser-native applications including a mathematical vector animation engine.
- Engineered Web Audio API applications for multi-thread sound synthesis, real-time generative audio, and responsive visualisation — fully client-side, zero dependencies.

Bare-Metal Infrastructure

- Maintain high-availability Ubuntu/Mint servers with low-level kernel tuning, custom systemd units, granular firewall policy, Nginx reverse proxy, and ZeroTier mesh networking. No managed-cloud dependency.
- GPU/driver resolution at Mesa level (crocus driver, Mesa package pinning, kernel parameter management) to sustain production stability across hardware generations.

Virtual World Engineering & LSL

- Architected physics-based micro-simulators, procedural and sacred-geometry builds, secure ledger-validation backends, complex state machines, and synchronised cross-server communication systems in LSL.

Technical SEO & Growth

- Implemented programmatic JSON-LD schema and semantic HTML5; maintained Page 1 Google visibility through Core Web Vitals optimisation, asset delivery tuning, and crawl-budget discipline across 30+ client properties.

Logistics Truck Manager — Software-Led Operations

[Amazon UK — Milton Keynes](#) · 2006 – 2008

High-volume fleet dispatch orchestration at a primary logistics hub; software engineering was the differentiating contribution.

£400k Annual Savings — Logistics Forecasting Framework

- Independently conceived, designed, and coded a custom automated data-analysis and bottleneck-prediction framework — eliminating manual spreadsheet pipelines and surfacing predictive capacity shortfalls before they became dispatch failures.
- Captured approximately £400,000 in recurring annual savings; framework adopted facility-wide and integrated into regional management playbooks.

Operational Excellence & Recognition

- Maintained 100% precision rating across complex dispatch schedules, transport audit trails, carrier reconciliations, and facility compliance.
- Awarded consecutive Amazon Kaizen honours for software-centric workspace-optimisation tools; safety-tracking and agile asset-deployment frameworks adopted across the regional network.

Inventory Control & QA Auditor

[Amazon UK — Milton Keynes](#) · 2005 – 2006

- Root-cause investigation and programmatic database reconciliation to identify and resolve warehouse tracking deficiencies before production impact.
- Top-tier internal ranking for statistical auditing reliability and processing speed.

Earlier Roles

- Credit Control Advisor — NTL, Luton (2005) · Ranked top performer every month; consistently exceeded collection targets through outbound negotiation and debt recovery.
- Technical Sales Executive — Currys, Luton (2001–2004) · Consistently exceeded targets; specialist in technical product consultation, objection handling, and credit agreement closure.

MENTORSHIP & KNOWLEDGE TRANSFER

- Amazon UK — Mentored junior engineers and operations staff on systems thinking, automation, and data-driven problem solving.
- Creative Technology — Ran workshops on music production, engineering, and creative workflow; built automated asset-rendering pipelines generating 500,000+ organic international views.

- Workplace Technical Training — Delivered 1:1 and small-group instruction across studio engineering, live sound, IT/web fundamentals, and front-end deployment workflows.

EDUCATION

GNVQ Business Studies (Foundation & Intermediate) Sixth Form Community College, Luton

GCSEs — Mathematics (B), English (C), Science (C) Lea Manor High School, Luton

British National — full, unrestricted right to work in the UK. References available on request.